South Mass Market — Data Science Presentation Script

For presenting to Remy — plain-English throughout, every workstream covered

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Plain-English Summary

This deck covers everything completed since July 6th for the South Mass Market & Canada team — eight connected pieces of work, not eight separate projects. Each slide follows the same structure: what problem existed, what we did, what we found, and what comes next.

The headline story: we built a reliable call forecast, corrected two assumptions the team had held for a while (autopay cancellations and seasonal call patterns), found that credit quality predicts who calls far better than bill size does, and used all of that to build a new tool that can flag customers heading toward a confusing or upsetting bill before they ever call in. Two pieces are brand new this week: a 6-month forecast that proves the seasonal pattern is real customer behavior (not just more or fewer customers), and the proactive bill-shock alert model.

One slide is specifically worth pausing on for Remy: the credit-quality finding on slide 9. Jonathan flagged this as directly relevant to the customer-quality conversation the team has been having — which customers cost more to serve over time. Take your time there.

How to Walk Through It, Slide by Slide

Below is what to actually say on each slide, written in plain language, assuming Remy has no prior context. For the workstream slides, it's broken into the same three-part story every time: what the problem was, what we did about it, and what we found.

Slide 1 — Title

Just introduce the scope in one sentence — no need for a problem/result structure here, it's just an opener.

Say something like:

“This covers everything the data science work has delivered for South Mass Market since July 6th — eight connected pieces of work I want to walk you through.”

Slide 2 — The Journey So Far

Explain simply that these eight pieces of work aren't random — each one led naturally into the next.

Say something like:

“Here's the shape of everything we've done. It's not eight separate projects — each piece actually led into the next one, ending with the newest tool we built, which I'll get to.”

Slide 3 — Baseline Forecasting Dataset

The Problem: Before this, the data we needed to build any kind of forecast lived in five different places, and someone had to run five separate steps by hand every time to pull it together.

What I Did: I combined all five of those steps into one single, automatic process. Now it produces one clean row of data for every single day, going all the way back to July 2022.

The Result: I checked the numbers this produced against a completely different piece of work we did separately, and they matched within three-tenths of one percent of each other — which tells us this data is trustworthy.

Say something like:

“This is the foundation everything else is built on. I took five scattered pieces of data and turned them into one reliable pipeline, and I proved it's accurate by checking it against a totally separate piece of work.”

Slide 4 — Call Volume Forecasting Model

The Problem: There was no real way to predict how many calls we'd get on a given day. Staffing was basically a guessing game, reacting to whatever came in.

What I Did: I built a model that predicts call volume based on the day of the week, then improved it using years of real history to make sure it reflects how customers actually behave today.

The Result: The forecast is now accurate to within about 8% of what actually happens — a reliable number the team can plan staffing around.

Say something like:

“Before this, staffing was a guessing game. Now we have a forecast that's accurate within a few percentage points of what actually happens.”

Slide 5 — Usage-Alert Targeting Model

The Problem: We have almost 650,000 customers. There was no way to know, out of everyone who calls about a confusing bill, which ones we should proactively reach out to first.

What I Did: I built a model that looks at a customer's credit standing, deposit status, and how often they've contacted us before, to flag the ones most likely to need help.

The Result: Out of nearly 25,000 people who called about a bill, this narrowed it down to a precise list of under 1,000 — the ones most worth reaching out to directly.

Say something like:

“Instead of guessing who might call confused about their bill, we can now pinpoint the roughly 900 customers most likely to — out of almost 650,000.”

Slide 6 — IVR Call Volume & Abandonment Analysis

The Problem: More customers were hanging up before reaching an agent, and leadership didn't have a clear explanation why.

What I Did: I broke the data down by day of the week to see exactly when this was happening.

The Result: It's concentrated on the Monday right after a holiday — a predictable pattern we can plan staffing around, not a sign that something is broken.

Say something like:

“Abandoned calls spiked recently, which looked concerning — but we traced it to a predictable pattern: the Monday after a holiday.”

Slide 7 — Autopay Removal Analysis

The Problem: The team believed that new customers were the main reason people cancel autopay — about 38% of cancellations, based on an old estimate.

What I Did: I checked that belief against real enrollment records, then dug into the actual reasons customers give when they call to cancel.

The Result: The real number is 15.2%, much lower than assumed. And we found three specific, different reasons people actually cancel: the payment date doesn't match their pay schedule, a surprise fee shows up on their very first bill, or they say they never agreed to autopay in the first place.

Say something like:

“We assumed new customers were driving most autopay cancellations — the real data shows that's not true, and pointed us to three specific, fixable reasons instead.”

Slide 8 — Forecast Refresh: Seasonality

The Problem: Our call forecast didn't account for the time of year at all, and the team assumed summer was our busiest season.

What I Did: I measured how call volume actually changes month to month, correcting for other factors so it's a fair comparison.

The Result: Calls are actually highest in January and February, not summer — the opposite of what the team expected.

Say something like:

“We assumed summer would be the busiest time for calls — turns out it's actually January and February.”

Slide 9 — Credit Quality Predicts Call Rate (SHOW REMY THIS — slow down here)

The Problem: Jonathan raised a concern: if we group customers by things like bill size and credit score, would each group even be big enough to trust the numbers?

What I Did: I split customers into groups by bill size and by credit quality, and counted how often each group actually calls in.

The Result: Every group turned out to have plenty of customers in it, so the numbers are trustworthy. And the real finding is this: at every bill size, customers with lower credit scores call in two-and-a-half to three times more often than customers with higher credit scores. Credit quality predicts calling behavior much more strongly than the size of the bill does.

Say something like:

“This is the one I want to spend a moment on. Credit quality predicts how often someone calls far more strongly than how high their bill is — and that's a real, measurable input into which customers we want to be bringing on.”

Slide 10 — Forecast Refresh: Call-Reason Breakdown

The Problem: We knew how many calls we were getting overall, but not which specific issues were actually driving that volume.

What I Did: I categorized calls from the last six months by the actual reason for the call.

The Result: Setting up a payment arrangement is the single biggest reason people call — about a quarter of all calls. That's the clearest opportunity to reduce call volume by making that easier for customers to do themselves.

Say something like:

“This confirms Payment Arrangement really is the single biggest reason people call — so that's where self-service investment would help the most.”

Slide 11 — 6-Month Forecast: Behavior, Not Population (New This Week)

The Problem: When we saw calls were higher in winter, the question was: is that because we simply have more customers in winter, or because people are genuinely calling more often?

What I Did: I looked at the number of active customers and the number of forecasted calls separately, month by month, for the next six months.

The Result: The number of customers barely changes at all across six months — less than a tenth of a percent. But the number of calls swings by 23%. Since the customer count is flat, that swing has to be real behavior, not just more people on the books.

Say something like:

“This is brand new this week. We proved the seasonal swing in calls is real behavior, not just more or fewer customers on the books.”

Slide 12 — Proactive Usage & Bill Shock Alert Model

The Problem: Up to now, everything we've built reacts after a customer has already called upset about a bill. There was nothing that catches the problem before it happens.

What I Did: I compared customers who called about a bill against similar customers who didn't, looking at their bill increase, credit score, and how long they've been a customer. Then I tested a few different combinations of rules for flagging someone in advance.

The Result: Every one of those factors is genuinely different between people who call and people who don't — confirming this is a real, usable signal. We now have a menu of options for how broadly to send alerts, ready for a decision on which one to use.

Say something like:

“This is the newest and most forward-looking piece. Instead of reacting after someone's already upset about their bill, we can now flag them before it happens.”

Slide 13 — Key Findings

This is the wrap-up — read the four rows as the headline takeaways of the entire deck, in plain terms.

Say something like:

“So across everything we've covered: the forecast is accurate and we know it reflects real behavior, credit quality is our strongest signal for who needs help, two things the team assumed turned out to be wrong, and we now have a tool that works proactively instead of just reacting.”

Slide 14 — Next Steps

Frame this as momentum, not a wish list — natural next steps building on what's already delivered.

Good moment to invite Remy's input directly on priorities.

Say something like:

“None of this requires new investment to keep going — but here's where it's headed next, and your input on priorities would help.”

Questions to Ask Remy

Framed for her role as SVP of Operations & Customer Experience — focused on staffing, customer experience, and operational priorities rather than the technical details.

1. Does the credit-quality finding on slide 9 change how you think about the customer-quality conversation, or does it confirm what operations has already been seeing?

2. From a customer experience standpoint, does the autopay finding (customers upset about draft timing and surprise fees) match what your team hears in complaints or escalations?

3. Should the bill-shock alert model be built with your customer experience team's input on what the actual customer message says, before it goes live?

4. Does the seasonal forecast — more calls in January/February — line up with how staffing has historically been planned for those months?

5. Is there a customer fairness or perception angle we should think through before using credit score as part of a targeting or alerting model?

6. Would you want the abandonment-rate finding (the holiday-Monday pattern) folded into how staffing gets planned around future holidays?

7. Which of these pieces of work would be most useful to bring to other markets or teams outside South Mass?

8. Would you want to see updates like this on a regular schedule, or only when there's a significant new finding?

9. Is there a specific customer experience or operational metric you track that you'd want us to try to move next?

10. Is there anything here that changes a decision you were already planning to make, or that you'd want more detail on before acting?

If She Asks You... (Anticipated Questions & Plain-English Answers)

Written for a non-technical audience — no data-science terms, focused on what it means for operations and customers.

Q: How do I know these numbers are actually right?

A: For each finding, we checked it against real historical records, not guesses. In a couple of cases, we actually caught our own mistakes first — an early version of one number came out looking way too extreme, and we found and fixed the reason before trusting it. So what's in this deck has already been double-checked.

Q: What does this actually mean for our customers, day to day?

A: A few things: customers who are heading toward a confusing or upsetting bill can now be reached before they ever have to call in frustrated. Customers calling about payment arrangements — our single biggest reason for calls — are the clearest opportunity to make that easier to do without calling at all. And we now understand why call volume spikes at certain times, so staffing can be planned around it instead of reacting to it.

Q: Is this going to change how we treat certain customers unfairly?

A: This isn't about treating anyone differently at the point of service — it's about understanding, after the fact, which customers are more likely to need extra help so we can reach out proactively and supportively. It's the same idea as reaching out before a bill surprises someone, not restricting anyone's access.

Q: Will this affect staffing or call center operations?

A: Yes, in a good way — we now have a forecast accurate enough to plan staffing around, including knowing that January and February are actually our busiest months, not summer as assumed. We also know the recent spike in abandoned calls was a predictable holiday pattern, not a sign of a real problem.

Q: How long did this take, and what did it cost us?

A: This is about three to four weeks of work on top of a full existing workload, using data and systems we already have. There was no new spend required to get to this point.

Q: Is any of this actually being used yet, or is it all still theoretical?

A: The call forecast and the autopay findings are ready to use today. The credit-quality work and the bill-shock alert model are validated and ready for the team to decide how to roll out next.

Q: What do you need from me to keep this going?

A: Mostly just prioritization — which of these should move into active use first — and in a couple of cases, help getting access to systems or people (like the software installation and the account-manager conversation already in progress with Jonathan).

Q: Why does this need to be a dedicated role instead of something done occasionally?

A: Because these findings only came to light by someone having the dedicated time to dig into the data. Without that, we'd likely still be operating on the old assumptions — that new customers drive most autopay cancellations, or that summer is our busiest season — both of which turned out to be wrong.

Q: How does this connect to the broader customer experience strategy?

A: Directly — it gives real numbers behind things that are usually judgment calls: which customers are more likely to struggle, which complaint is most common, and when we're most likely to be understaffed. That turns customer experience decisions into ones backed by data instead of instinct alone.